

```
import javax.swing.*;
```

```
import java.awt.*;
```

```
@SuppressWarnings("ALL")
```

```
public class numberGuessingGame extends JFrame {
```

```
    private final int randomNumber;
```

```
    private int turnsLeft;
```

```
    private final JTextField guessField;
```

```
    private final JButton guessButton;
```

```
    private final JTextArea resultArea;
```

```
    private boolean isGameEnded = false;
```

```
    public numberGuessingGame() {
```

```
        try {
```

```
            UIManager.setLookAndFeel(UIManager.getSystemLookAndFeelClassName());
```

```
        } catch (Exception e) {
```

```
            e.printStackTrace();
```

```
        }
```

```
        setTitle("Number Guessing Game");
```

```
        setSize(400, 200);
```

```
        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
```

```
        setLocationRelativeTo(null);
```

```
        randomNumber = (int) (Math.random() * 100) + 1;
```

```
        turnsLeft = 10;
```

```
        guessField = new JTextField(10);
```

```
        guessButton = new JButton("Guess");
```

```
        resultArea = new JTextArea();
```

```
        resultArea.setEditable(false);
```

```

// font size

Font largeFont = new Font("Arial", Font.PLAIN, 16);
guessField.setFont(largeFont);
guessButton.setFont(largeFont);
resultArea.setFont(largeFont);


// layout

setLayout(new BorderLayout());

JPanel panel = new JPanel();

JLabel label = new JLabel("Enter your guess: ");
label.setFont(largeFont);
panel.add(label);
panel.add(guessField);
panel.add(guessButton);
add(panel, BorderLayout.NORTH);
add(new JScrollPane(resultArea), BorderLayout.CENTER);


// Set up event handling
guessButton.addActionListener(e -> handleGuess());
}


private void handleGuess() {
    try {
        int userGuess = Integer.parseInt(guessField.getText());
        checkGuess(userGuess);
        turnsLeft--;

        if (turnsLeft == 0 || isGameEnded) {
            endGame();
        }
    }
}

```

```
    } catch (NumberFormatException ex) {  
        JOptionPane.showMessageDialog(this, "Please enter a valid number.");  
    }  
}
```

```
private void checkGuess(int userGuess) {  
    if (userGuess < randomNumber) {  
        resultArea.append("Your guess (" + userGuess + ") is too small.\n");  
    } else if (userGuess > randomNumber) {  
        resultArea.append("Your guess (" + userGuess + ") is too big.\n");  
    } else {  
        resultArea.append("Your guess (" + userGuess + ") is CORRECT!\n");  
        isGameEnded = true;  
    }  
}
```

```
    resultArea.append("Turns left: " + turnsLeft + "\n");  
    guessField.setText("");  
}
```

```
private void endGame() {  
    if (!isGameEnded) {  
        resultArea.append("Sorry, you ran out of turns. The correct number was: " + randomNumber  
+ "\n");  
    } else {  
        resultArea.append("Congratulations! You found the number in " + (10 - turnsLeft) + "  
turns.\n");  
    }  
}
```

```
    guessField.setEnabled(false);  
    guessButton.setEnabled(false);  
}
```

```
public static void main(String[] args) {  
    SwingUtilities.invokeLater(() -> new numberGuessingGame().setVisible(true));  
}  
}
```